

Table 1. Reconstructed ancestral olfactory receptor repertoires at the first three duplications, and the combinatorial code they support.

Node (Fig. 3)	Node ID	OR class	Human ORs descended	Ligands	Retained	Gained	Lost	Carboxylic acid	Aromatic ring
0	node_1	root (mixed)	433	21	—	—	—	0%	62%
1	node_2	Class I	62	21	4	17	17	48%	33%
2	node_3	Class II	371	8	7	1	14	0%	50%
3	node_4	Class I	55	23	12	11	9	83%	9%
4	node_5	Class I	7	21	13	8	8	57%	24%
5	node_6	Class II	370	27	7	20	1	0%	70%

Combinatorial code at each stage

Stage	Receptors	Odorants encoded	By 1 receptor	By ≥2 receptors	Combinatorial
S0	1 (node 0)	21	21	0	0%
S1	2 (nodes 1, 2)	27	25	2	7.4%
S2	3 (nodes 3, 4, 5)	53	36	17	32.1%

Repertoires. Ligand calls come from an isotonic density fill calibrated once on the 23,782 measured human OR–odorant pairs, applied separately to the ancestral matrix (cut $p \geq 0.8875$, density 2.26%). No experimental overlay is used: a reconstructed sequence has no measurements, so ancestors and extant receptors are compared prediction against prediction. Repertoire sizes are relative and should be read node against node, not as absolute binding counts.

Retained / gained / lost. Relative to the immediate parent node on the rooted human tree (node 0's parent is the hagfish outgroup and is not scored). Node 2's other daughter, the single receptor OR2AT4 (9606.A6NND4), sits on a long branch beside the outgroup and is excluded as a long-branch artefact; the Class II duplication is therefore read as node 2 → node 5.

Chemistry. Percentage of each node's ligands carrying the group (RDKit SMARTS). Enrichment is tested on the molecules **gained** along a branch, against the full 754-odorant panel (11.0% acid, 28.9% aromatic; Fisher exact, two-sided): branch 0 → 1 gains 10/17 acids (OR 13.0, $P = 2 \times 10^{-6}$) and no aromatic excess (OR 1.03, $P = 1.0$); branch 2 → 5 gains 15/20 aromatics (OR 7.9, $P = 2 \times 10^{-5}$) and no acids (0/20, $P = 0.15$).

Orthogonality. Each branch is summarised as the centroid of the molecules it gained minus the centroid of its parent's repertoire, in 13 panel-standardised RDKit descriptors. The two shift vectors are close to orthogonal (96.7°; cosine −0.12), against a null median of 33.3° from 2,000 random gain-sets of the same sizes — the observed angle exceeds all 2,000 draws (maximum 78.2°). Leave-one-descriptor-out spans 90–103°.

Combinatorial code. Stages are nested receptor sets: S0 = node 0; S1 = nodes 1 and 2; S2 = nodes 3, 4 and 5. An odorant is encoded if bound by ≥1 receptor of the stage and combinatorial if bound by ≥2. Counts are over the 754-odorant panel.